

1. GENERAL WELL INFORMATION

Well Name	Alpha-02	Well ID	W002
Field	Field_Alpha	Block	Block_1
Spud Date	14-06-2024	Rig Release Date	15-07-2024
Total Depth (TD)	3300 m	Rig Name	Rig Sagar-3
Well Type	Vertical Development	Trajectory	Vertical

2. FORMATION INFORMATION

Formation_X (unconsolidated sand-shale sequence) was encountered between **2820 m** and **3080 m**. This well is an offset to W001 within the same field block and exhibited similar loss behaviour at a comparable structural depth, along with an overpressure kick deeper in the section.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2500-2820	Formation_W	Shale, grey, firm	Normal drilling trend
2820-3010	Formation_X	Sand, friable, interbedded shale	Loss-prone; monitor returns
3010-3300	Formation_Z	Shale, dark grey, over-pressured	Kick risk; monitor gas trend

4. HISTORICAL DRILLING EVENTS

Event 1: Partial Mud Loss at approximately 2885 m

Observation: Gradual decrease in pit level was observed while drilling the sand streak of Formation_X, consistent with the loss event seen in W001.

Engineering Action: Pumps were slowed, LCM pill was spotted across the loss zone, and mud weight was reduced from 1.20 SG to 1.16 SG.

Outcome: Losses stabilized within 2 hours. Drilling resumed at reduced ROP through the remainder of the sand interval.

Event 2: Kick (Overpressure) at approximately 3200 m

Observation: Flow was observed on a connection with pit gain of 6 bbl; background gas increased from 40 to 210 units.

Engineering Action: Well was shut in using the BOP. Shut-in drill pipe and casing pressures were recorded, and kill mud weight was calculated and circulated using the Driller's Method.

Outcome: Well was successfully killed after two circulations. Mud weight was raised from 1.28 SG to 1.42 SG and drilling resumed under close monitoring.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
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Conductor	20	55	Surface
Surface	13-3/8	640	Surface
Intermediate	9-5/8	2400	1150
Production	7	3300	2350

6. RECOMMENDATION

Treat the Formation_X sand interval (2820-3010 m) as loss-prone based on repeated behaviour across Field_Alpha wells. Increase connection gas monitoring frequency below 3100 m and maintain kill-weight mud readily available ahead of the Formation_Z overpressure zone.

This Well Completion Report is a synthetically generated historical record produced for the NWIS prototype (W002). Not associated with an actual drilling operation.